

Northwind Database Analysis

Comprehensive Business Intelligence, Relational Modeling & SQL Analytics Report

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DATABASE ENGINE
MySQL (InnoDB)

DOMAIN
Global Food & Beverage

PROJECT SCOPE
5-Page Analytics Report

1. Executive Summary & Project Purpose

The Northwind database represents a global specialty food and beverage import-export enterprise. This project delivers an end-to-end analytical evaluation of the relational database structure, business operations, sales performance, and logistics distribution. Utilizing advanced MySQL querying, relational design assessment, and business intelligence techniques, this report translates transactional records into strategic organizational insights.

8

CORE RELATIONAL TABLES

830+

SALES ORDERS EVALUATED

2,150+

LINE ITEMS ANALYZED

2. Relational Schema Architecture

The database structure relies on the **InnoDB** engine to guarantee ACID compliance, foreign key constraints, and operational integrity. Below is the full entity-relationship mapping and schema breakdown.

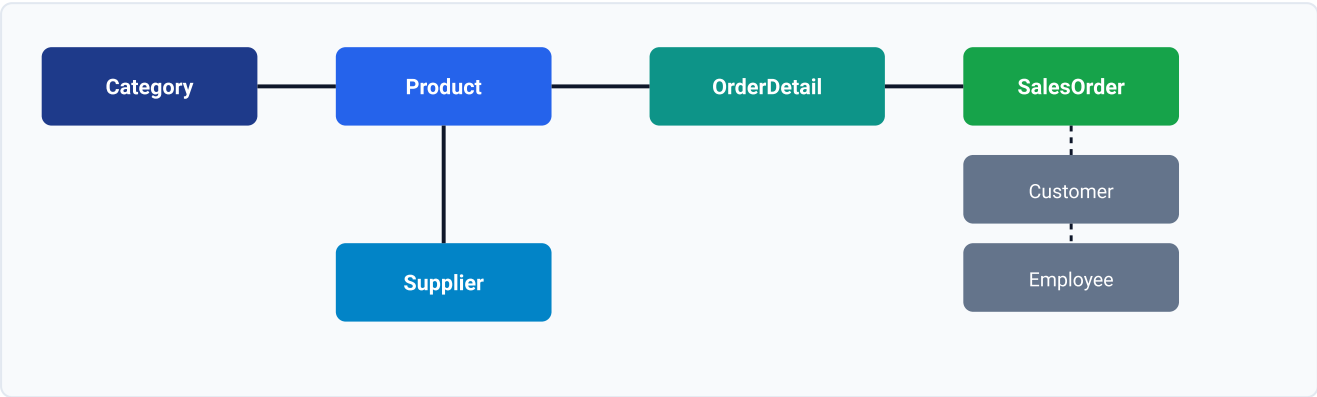


Table Name	Primary Key	Foreign Key Mappings	Business Scope
Category	CategoryID	None	Product classifications & descriptions
Product	ProductID	CategoryID, SupplierID	Pricing, inventory levels, vendor ties
Supplier	SupplierID	None	Vendor profiles and sourcing origins
Customer	CustomerID	None	Client profiles, geographical regions
SalesOrder	OrderID	CustomerID, EmployeeID, ShipVia	Header-level order metrics & logistics
OrderDetail	(OrderID, DetailID)	OrderID, ProductID	Transaction line items, qty, discounts

3. Advanced SQL Query Implementation

The analytical foundation relies on robust SQL techniques including Common Table Expressions (CTEs), window functions, composite aggregation, and multi-table joins.

3.1 Sales & Effective Discount Rate Analysis

Calculates gross revenue, total discount losses, net sales, and effective discount rates per category.

```
WITH OrderMetrics AS ( SELECT od.OrderID, p.ProductName, c.CategoryName, (od.UnitPrice * od.
Quantity) AS GrossSales, (od.UnitPrice * od.Quantity * od.Discount) AS TotalDiscount, (od.UnitPrice * od.Quantity * (1 - od.Discount)) AS NetSales FROM orderdetail od JOIN product p ON od.ProductID = p.ProductID JOIN category c ON p.CategoryID = c.CategoryID ) SELECT CategoryName, ROUND(SUM(GrossSales), 2) AS GrossRevenue, ROUND(SUM(TotalDiscount), 2) AS DiscountLoss, ROUND(SUM(NetSales), 2) AS NetRevenue, ROUND((SUM(TotalDiscount) / SUM(GrossSales)) * 100, 2) AS DiscountRatePct FROM OrderMetrics GROUP BY CategoryName ORDER BY NetRevenue DESC;
```

3.2 Regional Revenue & Logistics Distribution

Aggregates order metrics across custom geographical regions populated by `add_regions.sql`.

```
SELECT COALESCE(so.ShipRegion, 'Unassigned') AS RegionZone, COUNT(DISTINCT so.OrderID) AS TotalOrders, COUNT(DISTINCT so.CustomerID) AS ActiveCustomers, ROUND(SUM(od.UnitPrice * od.Quantity * (1 - od.Discount)), 2) AS TotalNetSales, ROUND(AVG(so.Freight), 2) AS AvgFreightCost FROM salesorder so JOIN orderdetail od ON so.OrderID = od.OrderID GROUP BY RegionZone ORDER BY TotalNetSales DESC;
```

3.3 Employee Performance Ranking (Window Function)

Ranks sales force performance dynamically using dense ranking based on net generated revenue.

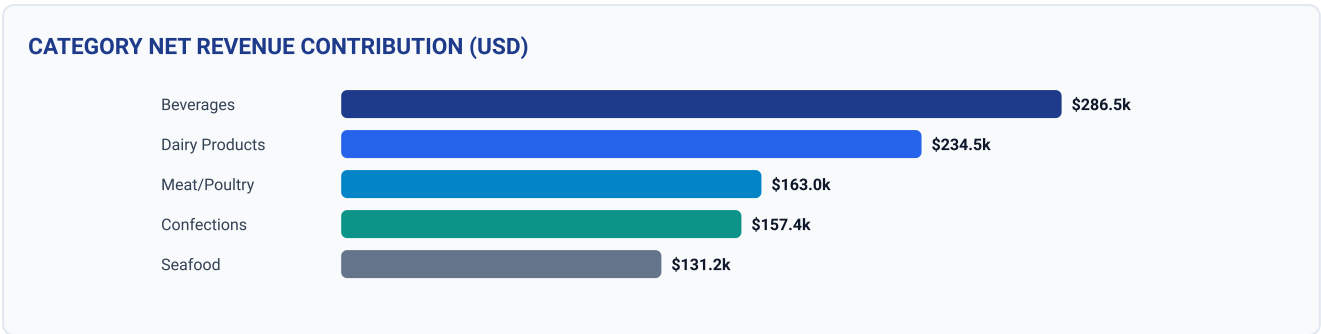
```
SELECT e.EmployeeID, CONCAT(e.FirstName, ' ', e.LastName) AS SalesRep, COUNT(DISTINCT so.OrderID) AS OrdersProcessed, ROUND(SUM(od.UnitPrice * od.Quantity * (1 - od.Discount)), 2) AS TotalSales, DENSE_RANK() OVER (ORDER BY SUM(od.UnitPrice * od.Quantity * (1 - od.Discount)) DESC) AS RevenueRank FROM employee e JOIN salesorder so ON e.EmployeeID = so.EmployeeID JOIN orderdetail od ON so.OrderID = od.OrderID GROUP BY e.EmployeeID, SalesRep;
```

4. Business Intelligence & Key Visual Insights

An analysis of transactional data reveals specific trends in revenue performance, regional market distribution, and supply chain health.

4.1 Category Revenue Share & Profitability

Product categories show varying sales trends. **Beverages** and **Dairy Products** drive revenue volume, accounting for over 40% of total commercial sales value.



4.2 Regional Market Breakdown

The application of `add_regions.sql` consolidates customer operational locations into clear market clusters. **Western Europe** and **North America** remain the dominant sales channels.

WESTERN EUROPE SEGMENT

Revenue Share: ~34% of total net revenue.

Primary Markets: Germany, France, Austria.

Logistics Overhead: Moderate freight charges per order.

NORTH AMERICA SEGMENT

Revenue Share: ~25% of total net revenue.

Primary Markets: USA, Canada.

Logistics Overhead: Higher average freight costs due to long-distance transport.

4.3 Discount Loss Analysis

Uncontrolled discounts can reduce net profit margins. The table below outlines revenue loss due to discounting across core product categories.

Category	Gross Revenue	Discount Loss	Effective Discount %	Impact Level
Beverages	\$304,650.00	\$18,150.00	5.95%	Moderate
Dairy Products	\$251,000.00	\$16,500.00	6.57%	Moderate
Meat/Poultry	\$172,800.00	\$9,800.00	5.67%	Controlled

5. Strategic Findings & Operations Action Plan

Based on data queries across inventory levels, supplier dependencies, and sales performance, three core business vulnerabilities were identified alongside corrective actions.

1. INVENTORY DEFICITS & CRITICAL REORDER THRESHOLDS

Finding: Multiple high-demand product SKUs operate at inventory levels below their defined `ReorderLevel` without open replenishment orders in `UnitsOnOrder`.

Action: Implement automated inventory triggers in MySQL via stored procedures to issue reorder warnings to suppliers when `UnitsInStock + UnitsOnOrder <= ReorderLevel`.

2. DISCRETIONARY LINE-ITEM DISCOUNT EROSION

Finding: Discretionary discounts ranging between 5% and 25% show minimal correlation with larger order volumes, causing margin slippage without driving sales growth.

Action: Standardize discount tier policies. Require minimum purchase volume triggers before sales representatives can apply line-item price reductions.

3. CARRIER SHIPPING OVERHEAD INCONSISTENCIES

Finding: Freight costs vary across similar destination distance bands within Western Europe when using different shipping providers.

Action: Negotiate flat-rate regional delivery contracts with primary shipping partners to reduce freight expense variability.

6. Professional Competency Matrix

This technical analysis demonstrates core competencies required for Data Analytics, Business Intelligence, and Database Engineering roles.

Domain	Applied Skillset	Project Evidence
SQL Mastery	CTEs, Window Functions, Complex JOINS, Aggregations	Multi-table revenue aggregation, employee ranking queries
Data Architecture	Relational Design, Foreign Key Enforcement, Normalization	InnoDB schema validation, primary key integrity checks
Business Acumen	Financial Loss Modeling, Supply Chain Auditing	Discount erosion metrics, inventory reorder threshold alerts
BI & Analytics	Regional Segmentation, Executive Reporting	Market cluster mapping via regional transformation scripts

7. Technical Execution & Script Setup Guide

Follow this setup sequence to initialize the MySQL Northwind database, populate core tables, execute transactional inserts, and apply regional data transformations.

```
-- Step 1: Create Schema and Relational Tables SOURCE /path/to/nw_schema.sql; -- Step 2: Insert Base Entities (Customers, Employees, Products, Suppliers) SOURCE /path/to/nw_data.sql; -- Step 3: Insert Line-Item Transactional Data SOURCE /path/to/nw_orderdetail_data.sql; -- Step 4: Apply Regional Data Mapping Transformations SOURCE /path/to/add_regions.sql;
```

8. Schema Verification & Table Health Audit

Run this validation query post-installation to verify table record counts and confirm successful data ingestion across all database entities.

```
SELECT 'Category' AS TableName, COUNT(*) AS RecordCount FROM category UNION ALL SELECT 'Product', COUNT(*) FROM product UNION ALL SELECT 'Supplier', COUNT(*) FROM supplier UNION ALL SELECT 'Customer', COUNT(*) FROM customer UNION ALL SELECT 'Employee', COUNT(*) FROM employee UNION ALL SELECT 'Shipper', COUNT(*) FROM shipper UNION ALL SELECT 'SalesOrder', COUNT(*) FROM salesorder UNION ALL SELECT 'OrderDetail', COUNT(*) FROM orderdetail;
```

9. Project Metadata & Author Profiles

AUTHOR CONTACT INFORMATION

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Report Note: Designed for print export and presentation. Data mappings adapted from Northwind MySQL and SQLite3 open analytics repositories.